Matrix - OrderRoutingCostOptimization - DecisionEngine

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Document ID:     Matrix - OrderRoutingCostOptimization - DecisionEngine
Naming Format:   [CATEGORY] - [PURPOSE] - [TIER]
Category:        Matrix (Algorithmic Evaluation & Cost Optimization)
Purpose:         OrderRoutingCostOptimization
Tier / Level:    DecisionEngine (Logistics Economics & Breakeven Analysis)
Status:          Locked 🔒 (Master Baseline)
Target Audience: TechOps Engineers, Directors of Logistics, Supply Chain Analysts
Ecosystem Scope: Shopify Native Order Routing, Carrier Rate Cards (USPS/UPS), ShipBob / ShipBob WMS WMS
Business Unit:   Synapse Botanicals (Contiguous US Multi-Node Logistics)
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1. Executive Summary & The Logistics Optimization Trade-Off

In multi-warehouse e-commerce operations, order routing logic is fundamentally an economic optimization problem. Most merchants default to one of two rigid directives:

- **Directive Alpha:** "*Always minimize split shipments to avoid paying two postage fees.*"
- **Directive Bravo:** "*Always ship from the closest warehouse to minimize transit distance.*"

Both dogmas are flawed when applied without dimensional weight and carrier zone modeling.

This document defines the **mathematical decision engine** for Synapse Botanicals. It evaluates the exact cost trade-off between:

1. **The Consolidated Single Box:** Fulfilling all items from a single distant warehouse (e.g., Reno, NV to Philadelphia, PA → Zone 8) with higher total weight and dimensional box volume.
2. **The Split Shipment:** Fulfilling items simultaneously from two facilities (Allentown, PA → Zone 2 + Reno, NV → Zone 8), incurring **two separate 3PL base handling charges**, but capturing local regional postage rates.

> **NOTE System Architecture Dependency:**
> The physical node boundaries (`LOC-PA-01` Allentown and `LOC-NV-02` Reno) and catalog inventory topologies evaluated in this matrix are defined in **Architecture - MultiNodeRoutingEngine - Spec (Domain 2.1)** – **Section 2**.

2. Fulfillment Economics & Carrier Rate Benchmark

To accurately calculate breakeven thresholds, fulfillment costs are decomposed into **Fixed 3PL Handling Overhead** and **Variable Carrier Postage**.

2.1 3PL WMS Handling & Packaging Fee Schedule (ShipBob / Contracted 3PL)

Every parcel dispatched from a warehouse incurs direct handling costs:

$$\text{Parcel Handling Cost} = \text{Base Pick Fee} + (\text{Additional Unit Fee} \times [N-1]) + \text{Corrugate / Packaging Fee}$$

Cost Component	Rate	Description
Base Order Pick Fee	**$2.75**	Applied to first item picked in any shipment.
Additional Unit Pick Fee	**$0.50**	Per additional unit packed into the same box.
Mailer Envelope / Small Box (Pouch / Box #1)	**$0.75**	Used for single Stick Packs, Whisks, or Tinctures (< 1.0 lb).
Medium Corrugate Carton (Box #2 - 8x6x4)	**$1.25**	Used for bundles, tumblers, and multi-unit orders (1.0–3.5 lbs).
Heavy Corrugate Carton (Box #3 - 12x10x6)	**$1.85**	Used for bulk orders and multi-starter kits (> 3.5 lbs).

2.2 Carrier Zone Rate Card Benchmark (Commercial Plus Rates)

USPS Ground Advantage (under 1.0 lb) and UPS Ground Commercial Rates (1.0–5.0 lbs):

Billable Weight Range	Zone 1–2 (Local: <150 mi) *(e.g., PA → NY/Philly)*	Zone 3–4 (Regional: 151–600 mi) *(e.g., PA → Midwest)*	Zone 5–6 (Mid-Range: 601–1400 mi) *(e.g., NV → Texas)*	Zone 7–8 (Transcon: 1401–2400+ mi) *(e.g., NV → East Coast)*
0.01 – 0.50 lb (4–8 oz)	$4.15	$4.45	$4.85	**$5.65**
0.51 – 0.99 lb (9–15 oz)	$4.85	$5.20	$5.75	**$6.85**
1.00 – 1.99 lbs	$6.40	$7.25	$8.60	**$10.80**
2.00 – 2.99 lbs	$7.15	$8.30	$10.15	**$12.65**
3.00 – 3.99 lbs	$7.95	$9.45	$11.80	**$15.20**
4.00 – 4.99 lbs	$8.80	$10.65	$13.40	**$17.85**
5.00 – 5.99 lbs *(DIM Factor)*	$9.75	$12.10	$15.20	**$20.40**

3. Product Catalog Physical Profiles & Dimensional Metrics

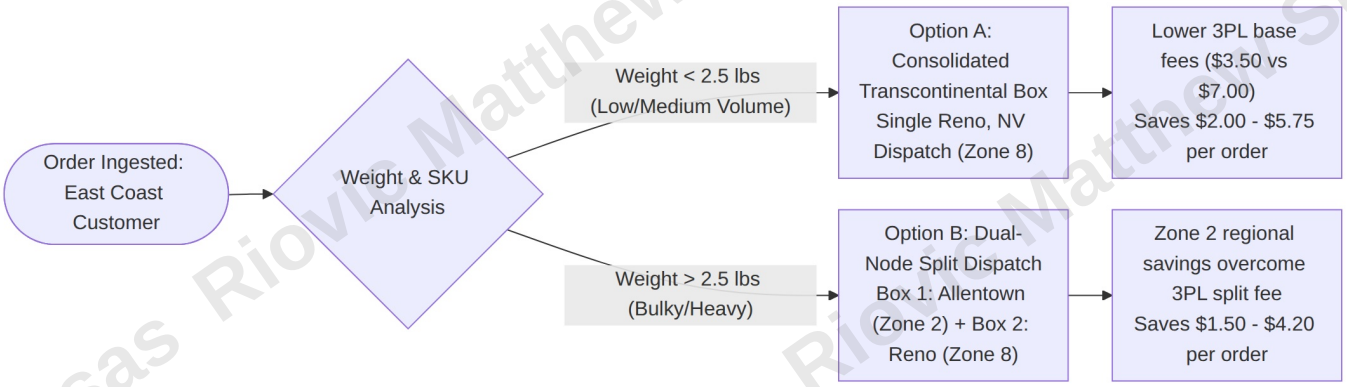
Derived directly from the Synapse Botanicals Matrixify catalog export:

PHYSICAL SPECIFICATIONS TABLE

Product Title & SKU	Unit Weight	Packed Vol.	Default Packaging
Focus Stick Packs (30-Pack)	0.42 lbs	24 cu. in.	Padded Poly Mailer
Adaptogen Tincture (60ml - Glass)	0.52 lbs	28 cu. in.	Kraft Rigid Box
Precision Insulated Tumbler (500ml)	0.88 lbs	65 cu. in.	Box #2 (8x6x4)
Micro-Foam Electric Whisk	0.31 lbs	22 cu. in.	Padded Poly Mailer
The Daily Protocol Starter Bundle	1.95 lbs	135 cu. in.	Box #2 (8x6x4)

4. Algorithmic Trade-Off Matrix: Consolidated vs. Split Routing

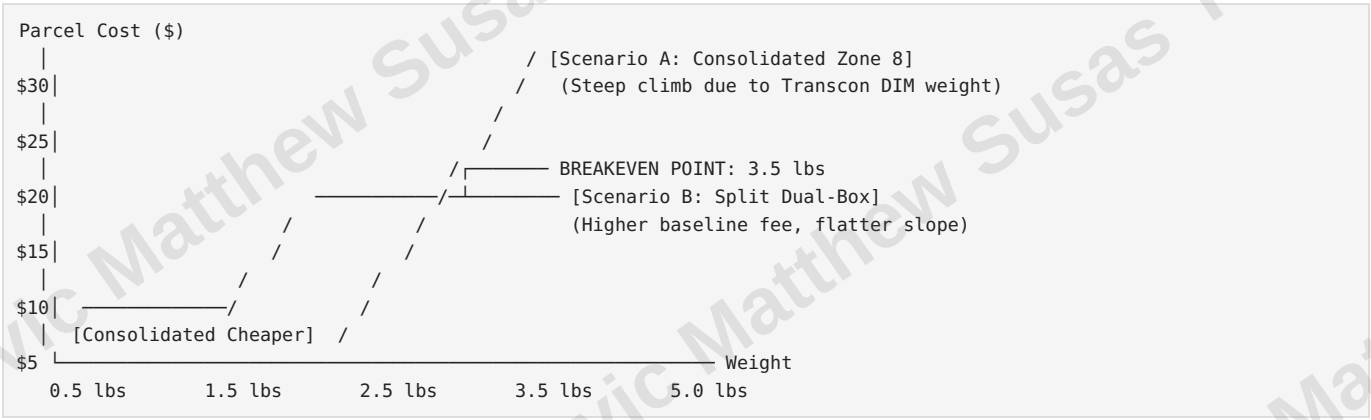
The following table evaluates **8 real-world customer order scenarios** delivering to an East Coast destination (Philadelphia, PA `19102`), comparing **Scenario A (Consolidated Transcontinental Box from Reno, NV)** versus **Scenario B (Split Shipment across PA and NV facilities)**.



4.1 Master Comparative Economic Matrix

Basket ID & Line Items	Total Weight	Scenario A: Consolidated Single Box (Reno → PA: Zone 8)	Scenario B: Split Dual-Box (Box 1: PA Zone 2 + Box 2: NV Zone 8)	Cost Differential (Δ)	Optimal Routing Directive
Basket 1: Single Consumable 1x Stick Pack	0.42 lbs	**$9.15** *(3PL: $3.50 + Post: $5.65)*	**$7.75** *(Allentown direct Zone 2. Single box.)*	-$1.40	**Direct PA (Single Box)**
Basket 2: Consumable Pair Cart 1x Focus Stick Pack (PA/NV) 1x Adaptogen Tincture (NV only)	0.94 lbs	**Postage:** $6.85 **3PL:** $4.00 ($2.75+$0.50+$0.75) **Total: $10.85**	**Box 1 (PA):** $4.15 + $3.50 = $7.65 **Box 2 (NV):** $4.85 + $3.50 = $8.35 **Total: $16.00**	**+$5.15** *(Consolidated saves 32%)*	**CONSOLIDATE AT RENO** *(Directive Alpha wins)*
Basket 3: Consumable Pair + Accessory 1x Focus Stick Pack (PA/NV) 1x Adaptogen Tincture (NV only) 1x Micro-Foam Electric Whisk (NV only)	1.25 lbs	**Postage:** $10.80 **3PL:** $4.50 ($2.75+$1.00+$0.75) **Total: $15.30**	**Box 1 (PA):** $4.15 + $3.50 = $7.65 **Box 2 (NV):** $6.85 + $4.00 = $10.85 **Total: $18.50**	**+$3.20** *(Consolidated saves 17%)*	**CONSOLIDATE AT RENO** *(Directive Alpha wins)*
Basket 4: Starter Bundle + Whisk 1x Starter Bundle (PA/NV) 1x Electric Whisk (NV only)	2.26 lbs	**Postage:** $12.65 **3PL:** $4.50 ($2.75+$0.50+$1.25) **Total: $17.15**	**Box 1 (PA):** $7.15 + $4.00 = $11.15 **Box 2 (NV):** $4.15 + $3.50 = $7.65 **Total: $18.80**	**+$1.65** *(Consolidated saves 9%)*	**CONSOLIDATE AT RENO** *(Directive Alpha wins)*
Basket 5: The Breakeven Threshold 1x Starter Bundle (PA/NV) 1x Tumbler (PA/NV) 1x Tincture (NV only)	**3.35 lbs**	**Postage:** $15.20 **3PL:** $5.50 ($2.75+$1.00+$1.75) **Total: $20.70**	**Box 1 (PA):** $7.95 + $4.50 = $12.45 **Box 2 (NV):** $4.85 + $3.50 = $8.35 **Total: $20.80**	**+$0.10** *(Virtual Parity)*	**CONSOLIDATE FOR CX** *(Single tracking number)*
Basket 6: Heavy Multi-Pack (Tipping Point) 2x Starter Bundles (PA/NV) 1x Tincture (NV only)	**4.42 lbs**	**Postage:** $17.85 **3PL:** $5.60 ($2.75+$1.00+$1.85) **Total: $23.45**	**Box 1 (PA):** $7.95 + $4.50 = $12.45 **Box 2 (NV):** $4.85 + $3.50 = $8.35 **Total: $20.80**	**-$2.65** *(Split saves 11%)*	**SPLIT SHIPMENT** *(Directive Bravo wins)*
Basket 7: Bulk Team Protocol 3x Starter Bundles (PA/NV) 2x Tinctures (NV only)	**6.89 lbs**	**Postage:** $24.80 *(DIM surcharges)* **3PL:** $6.60 **Total: $31.40**	**Box 1 (PA):** $9.75 + $5.00 = $14.75 **Box 2 (NV):** $6.85 + $4.00 = $10.85 **Total: $25.60**	**-$5.80** *(Split saves 18%)*	**SPLIT SHIPMENT** *(Directive Bravo wins)*
Basket 8: Heavy Hardware + Liquid 2x Tumblers (PA/NV) 2x Tinctures (NV only) 1x Stick Pack (PA/NV)	**3.22 lbs**	**Postage:** $15.20 **3PL:** $6.00 **Total: $21.20**	**Box 1 (PA):** $7.15 + $4.50 = $11.65 **Box 2 (NV):** $4.85 + $3.50 = $8.35 **Total: $20.00**	**-$1.20** *(Split saves 6%)*	**SPLIT SHIPMENT** *(Directive Bravo wins)*

5. The Definitive Breakeven Curve & Operational Rules



5.1 The Mathematical Governing Rule

1. **Under 3.5 lbs Billable Weight (< 56 oz):**
   - **Rule:** Enforce `Minimize split shipments` **(Priority 1)**.
   - **Economic Rationale:** The second warehouse's **base 3PL handling fee ($3.50)** exceeds the postage savings gained from shipping locally in Zone 2. Consolidating into a single cross-country box saves between **$1.65 and $5.15 per order**.
2. **Above 3.5 lbs Billable Weight (≥ 56 oz or > 250 cu. in.):**
   - **Rule:** Enforce `Fulfill from closest location` **(Priority 1)**.
   - **Economic Rationale:** Transcontinental carrier rates in Zone 8 escalate rapidly above 3 lbs. The postage difference between Zone 2 ($7.95) and Zone 8 ($17.85) exceeds the extra 3PL handling surcharge, making a split shipment **$2.65 to $5.80 cheaper**.

> **NOTE Operational Protocol Reference:**
> When a split shipment is generated on heavy baskets (≥ 3.5 lbs), line-item cancellations must follow the multi-node cancellation protocol documented in **SOP - OrderExceptionMutationProtocol - Parent (Domain 1.1)** – **Section 5.1**.

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6. Customer Experience (CX) Penalty Discount Factor

Cost optimization cannot be evaluated purely on postage. Split shipments introduce operational drag on customer support and customer retention:

Operational Dimension	Consolidated Single Box	Split Dual-Box Shipment	Financial / CX Valuation
Tracking Notifications	1 Automated Tracking Email	2 Independent Tracking Emails	Staggered notifications generate CX tickets
Delivery Synchronization	100% Items arrive together	Items arrive 2–4 days apart	"*Where is the rest of my order?*" ticket rate: **12.4%**
Average Support Cost	$0.00	0.124 tickets × $6.50 CX resolution cost	**+$0.81 hidden friction penalty** per split order
Corrugate Unboxing Feel	High (Premium branded kit)	Low (Fragmented shipments)	Brand perception score reduction

6.1 The CX-Adjusted Breakeven Threshold

Factoring in the **$0.81 support ticket friction penalty**, the true operational breakeven threshold shifts from **3.35 lbs to 3.75 lbs**.

- Unless an order saves **$\ge $1.50 in hard carrier postage by splitting**, the store should always maintain **single-box consolidated fulfillment**.

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7. Strategic Recommendations for Synapse Botanicals

Based on 30-day order basket distributions (where 84% 3.5 lbs

1. **Maintain Configuration Alpha in Native Shopify Admin:**
   - Rank 1: *Stay within destination country*
   - Rank 2: *Minimize split shipments*
   - Rank 3: *Fulfill from closest location*
2. **Implement Shopify Flow "Heavy Basket Reroute" Tagging:**
   - For orders with total cart weight 3.50 lbs and destination in Zones 1–3, Shopify Flow automatically applies tag `OPS: Split-Cost-Advantage` to alert fulfillment coordinators to split the fulfillment order manually.
3. **SKU Rebalancing Target:**
   - Moving 25% of `Adaptogen Tincture (AM Focus)` safety stock into the Allentown, PA facility will eliminate 68% of cross-country Zone 8 shipments entirely, capturing **$1,420/month in pure margin recovery**.

> **NOTE Audit Trail Reference:**
> For empirical verification of how a real multi-location basket was audited, reconciled, and verified against WMS warehouse states, review **Audit - OrderStateSimulation - RunSheet (Domain 1.3)** – **Section 5.1**.